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Abstract

The present invention relates to a trolling motor and related system thereof. More particularly, a system designed to address the limitation of existing trolling motor applications. The trolling motor system includes a control and deployment system. The remote deployment is designed for rearward propulsion, with no reference for steering as the motors are in a fixed state. The present invention also implements automatic deploying and stowing trolling motors, and brackets for ease of deployment and stow. The control system can be positioned at the row of a boat or a remote device and designed for various commands for the ease of use for an operator.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention relates to trolling motors and, more particularly, to a system designed to address the limitations of existing trolling motor applications. It includes remote deployment with rearward propulsion, with no reference for steering, as the motors are in a fixed state.

[0002] Trolling motors are mounted either at the bow or stern of boats and are designed to provide quiet and efficient propulsion, allowing anglers to move quietly through the water without disturbing fish. They are controlled by a hand-held device or foot pedals, enabling the operator to adjust speed, direction, and other settings as needed.

SUMMARY OF THE INVENTION

[0003] In view of the foregoing background, the present invention offers a new and novel concept that users are able to use forward facing sonar with ease. The rear mounted trolling motors reduces the chance of spooking/disturbing the fish in front of you. This is very helpful when fishing with forward facing sonar.

[0004] The present invention also implements automatic deploying and stowing trolling motors, and brackets for ease of deploy and stow. All the components can be changed to accommodate various types of boats, motors, and power sources.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0005] FIG. 1 is a perspective view of Power Box.

[0006] FIG. 2 is a perspective view of Center Bracket.

[0007] FIG. 3 is a perspective view of Raptor Bracket.

[0008] FIG. 4 is a perspective view of Stop button.

[0009] FIG. 5 is a perspective view of Complete Bracket.

[0010] FIG. 6 is a perspective view of Complete Bracket stowed.

[0011] FIG. 7 is a top view of Control button.

[0012] FIG. 8 is a perspective view of Control button.

[0013] FIG. 9 is a top view of Foot Pedal button.

[0014] FIG. 10 is a perspective view of Foot Pedal button.

DETAILED DESCRIPTION OF THE INVENTION

[0015] The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiment set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

[0016] FIG. 1 illustrates the power box 1 enclosing an electronic controller to control the heading and speed of the trolling motors. The electronic controller is to receive wireless input signals 3 from wireless devices or to receive wired signals 2 from wired devices. The devices such as the control button 9, 10, or the foot pedal button 11, 12 can send signals wirelessly or through wires connecting the devices to the controller in the power box 1. The devices have an integrated wireless system such as wireless, Bluetooth or other similar system to send wireless signals to the electronic controller.

[0017] A foot pedal assembly can be configured to mechanically link it to the controlling unit. A base plate of the foot pedal 11, 12 is attached to the floor through bolting or some other means. Attachment hardware such as brackets 7 or 8 are used for attaching mechanical cable to the trolling motor shaft. The stop button 6 is connected to the trolling motor and the boat's battery by electrical cable.

[0018] The trolling motors come in various sizes and power ratings, offering different levels of thrust and battery power to suit different boat sizes and usage scenarios. They are popular among fishermen for their ability to maneuver silently and precisely in various fishing environments, such as lakes, rivers, and shallow waters, without relying on the main engine.

[0019] Such a system requires a mechanism to combat wind and current to maintain desired speed and path so that anglers can focus on casting instead of steering. Constant adjustment must be made to hold a position. Frequent and smaller adjustment improves positioning for use of forward facing sonar.

[0020] Previously, front mounted trolling motors would be spun around to slow down, stop, wobble, and back up. Recently, some companies have started selling systems to mount on very expensive shallow water anchors.

[0021] The present invention comprises a set of rear mounted trolling motors that are deployed through an actuator from the front of the boat via a button on a control pedal. These trolling motors are speed/thrust controlled by a dial also integrated in to control pedal. These trolling motors allow a boat to slow, stop, and back up in the water with a greatly reduced chance of spooking/disturbing the fish in front of you. This is very helpful when fishing with forward facing sonar.

[0022] The present invention provides a foot pedal as in FIG. 9 and FIG. 10. to provide hands-free operation of the boat, freeing up both hands for casting, reeling. The foot pedal includes a plurality of switches for commanding operation of a trolling motor.

[0023] The present invention provides remote control for the operation and deployment of trolling motors. Transmitter of the remote controller sends wired or wireless signals to the receivers of trolling motors and motor deployment circuits.

[0024] The present invention will eliminate the need for the expensive shallow water anchors. A well-coordinated pair of trolling motors eliminates the need. These rear mounted trolling motors have the convenience of control from the front of the boat. This also adds speed/thrust control to the motors so they can adapt to the weather conditions that are always changing.

[0025] Unlike many existing trolling motor systems that are based upon forward propulsion and directability, the present invention is for slowing and stopping of a watercraft and the possibility of reverse propulsion.

[0026] The present invention allows deployment through electrical signal to take place from anywhere in the boat instead of being limited by a predetermined length of cable. The present invention covers the deployment brackets along with electric and gas charged deployment device, and its associated method. This design will only require minimum modifications, such as a few bolts, depending on the mounting option selected.

[0027] One skilled in the art, after being exposed to the teachings provided in the preceding descriptions and associated drawings, will likely conceive various modifications and alternative embodiments of the invention. Hence, it is important to note that the invention is not limited to the disclosed specific embodiments, and that modifications and alternative embodiments are intended to be encompassed within the scope of the appended claims.

[0028] Thus, the scope of the present disclosure is to be determined by the broadest permissible interpretation to the maximum extent allowed by law, of the following claims, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. A trolling motor control and deployment system, comprising: a pair of trolling motors; trolling motor control apparatus; trolling motor deployment apparatus; and trolling motor deployment bracket.
2. The pair of trolling motors of claim 1 is mounted at the stern of a boat.
3. The trolling motor control apparatus of claim 1 is positioned at the bow of a boat, or on a remote

device.

- 4.** The trolling motor of claim 1 comprising: a motor with a propeller; and a motor controller.
 - 5.** The motor controller of claim 4 comprises: means for commanding the operation of the motor; and means to receive and decode commands transmitted by remote devices.
 - 6.** The operation of the motor of claim 5 includes speed control, direction control, and other settings as needed.
 - 7.** The remote devices of claim 5 can be either a hand-held device or a foot pedal.
 - 8.** The transmission means of claim 5 can be either wired means or wireless means.
 - 9.** The trolling motor deployment apparatus of claim 1 is charged with electric or gas.
 - 10.** The deployment brackets of claim 1 allows the trolling motors of claim 1 to be deployed or stowed.
 - 11.** The method of deploying the trolling motors of claim 1 comprises steps of: releasing safety lock; starting the deployment apparatus of claim 1; and sliding the trolling motors of claim 1 off.
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